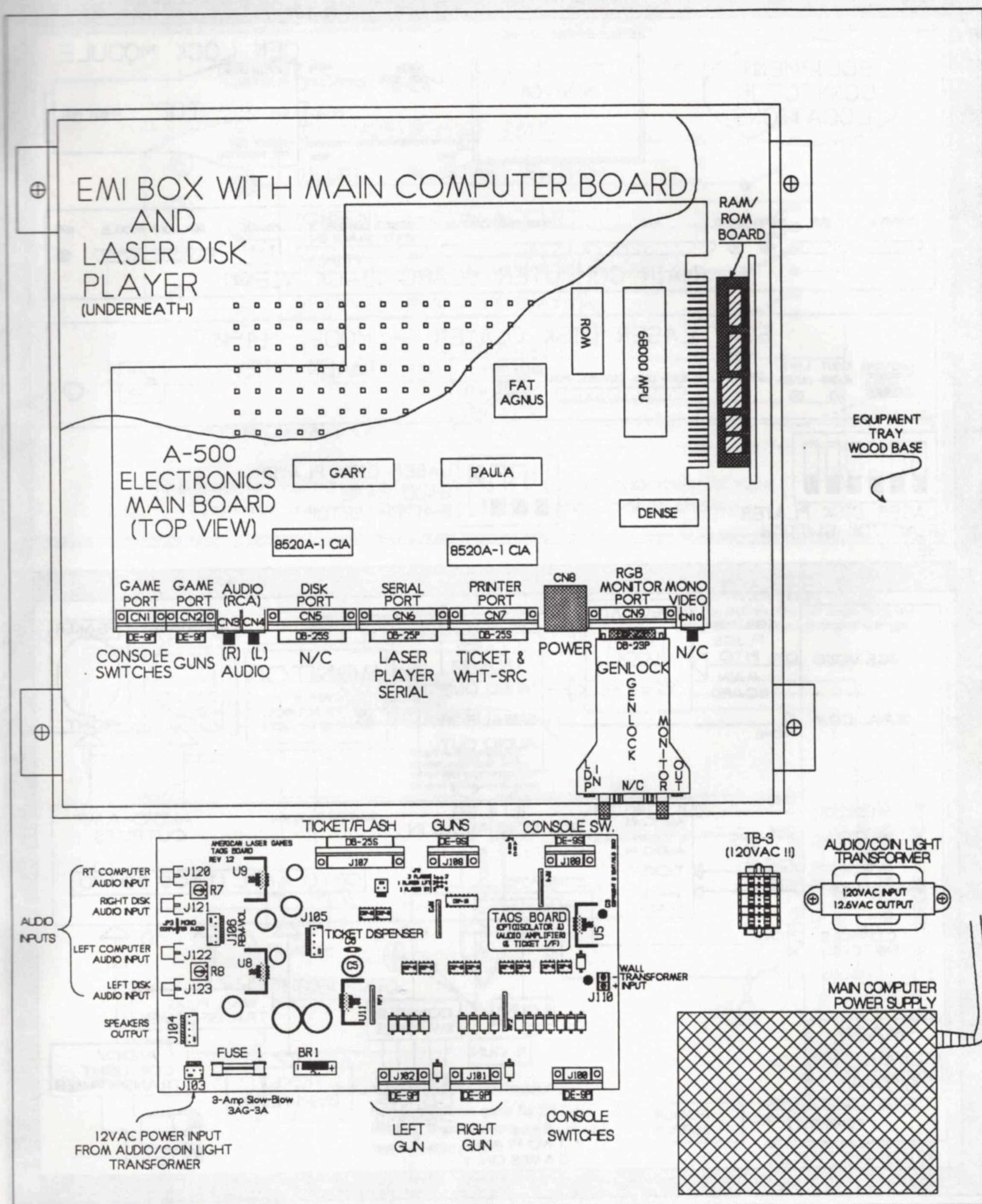
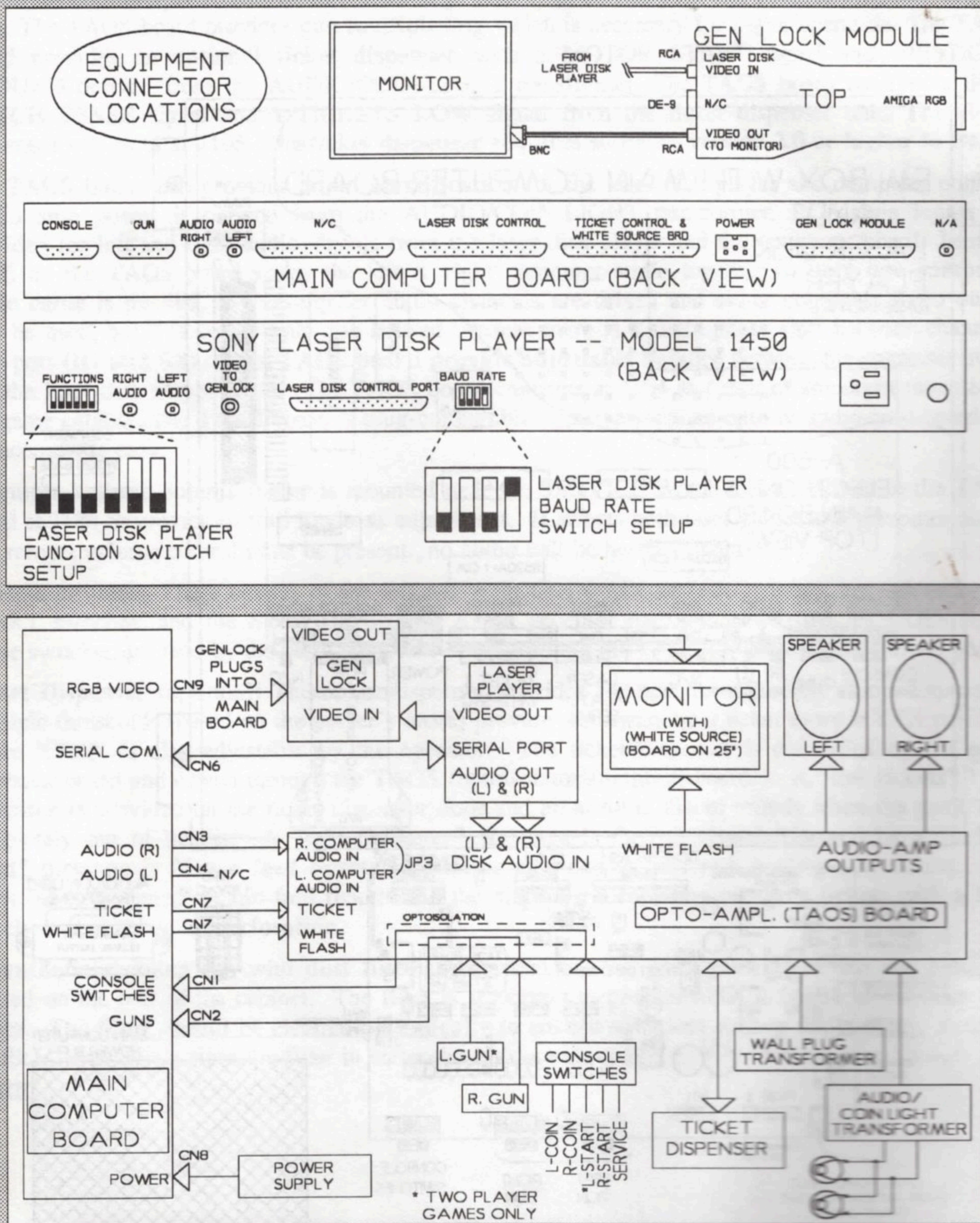


COMPONENT & CONNECTOR LOCATIONS IN EQUIPMENT TRAY

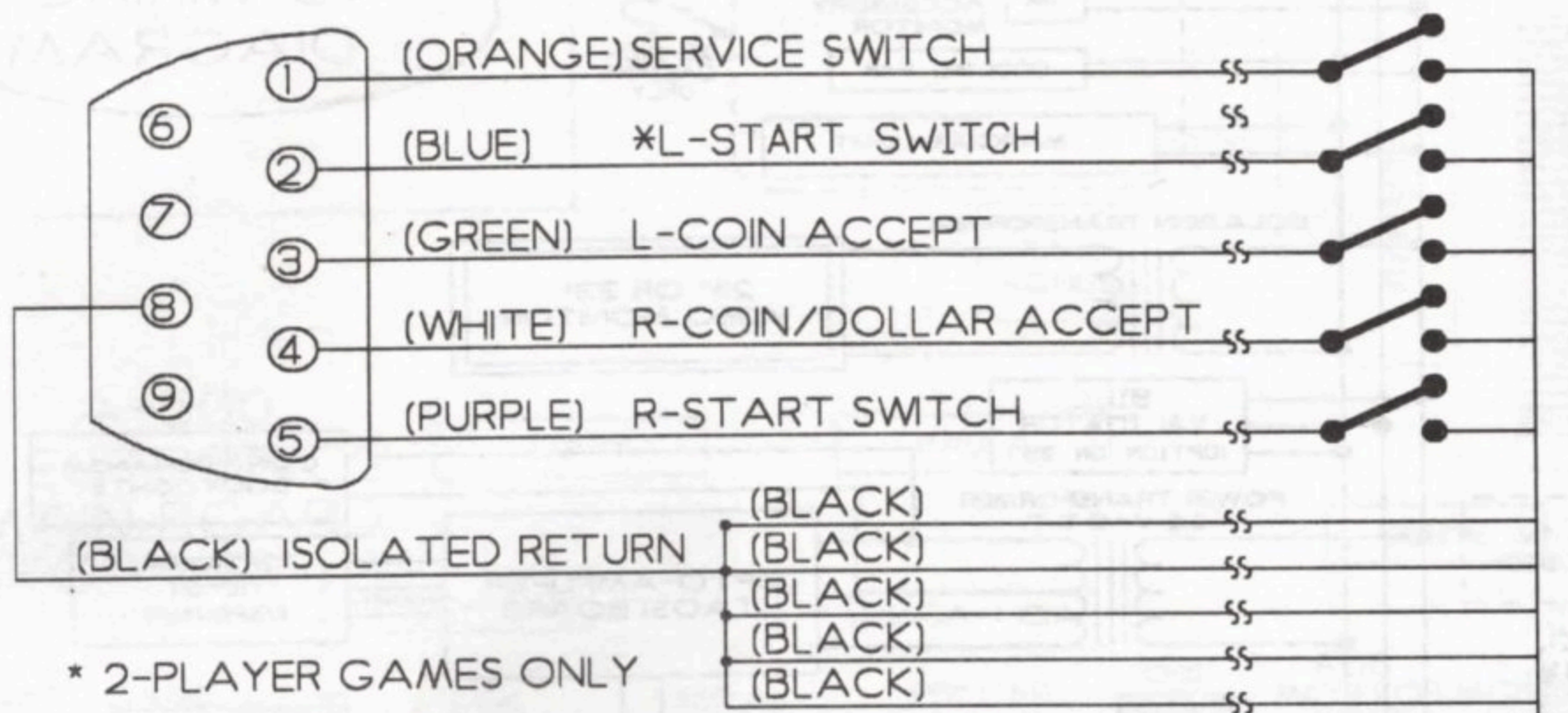


AMERICAN LASER GAMES



AMERICAN LASER GAMES

Console Switch Cable

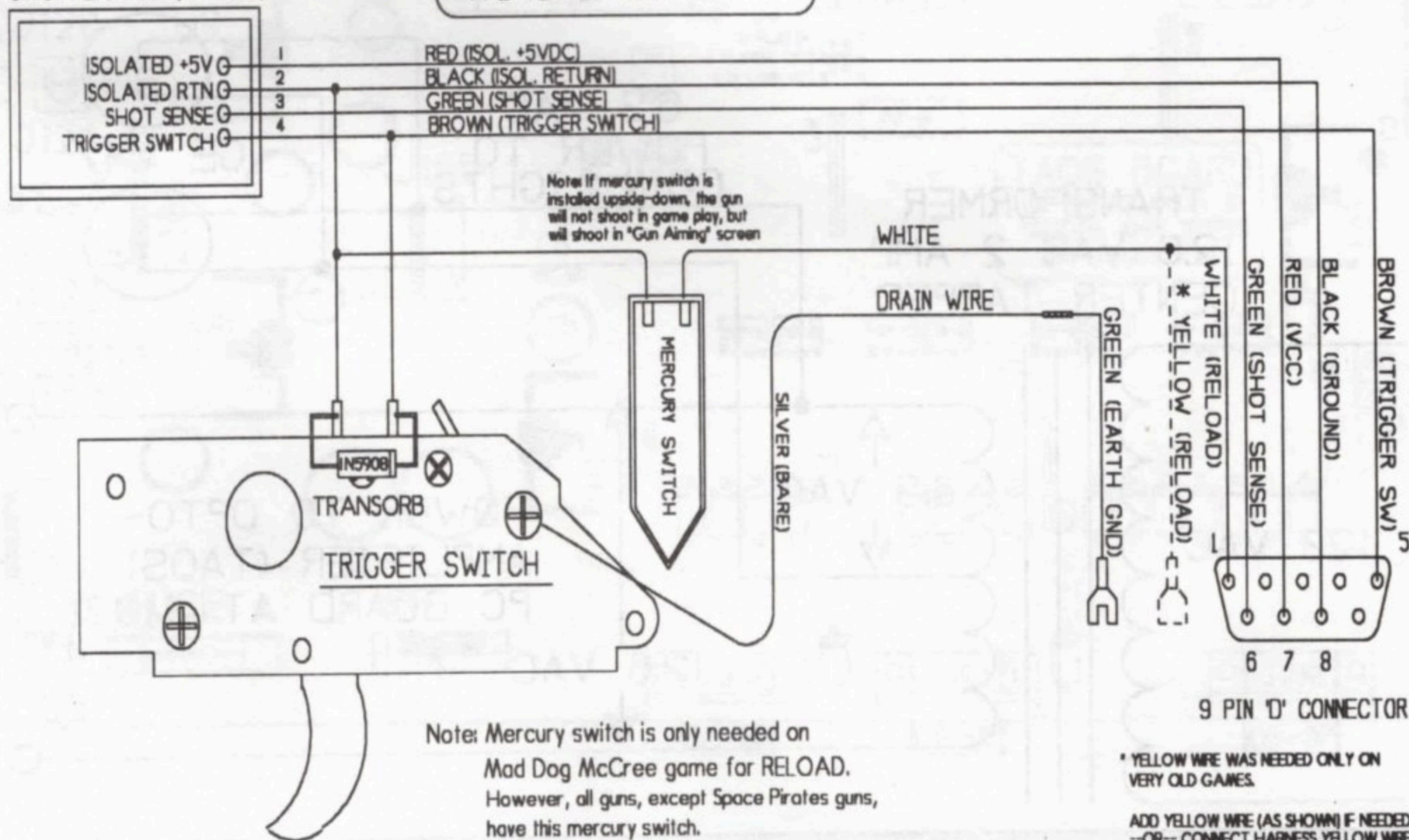


The cabinet harness connector is a 9 pin 'D' sub-mini female
The rear view (wiring side) of the connector is shown

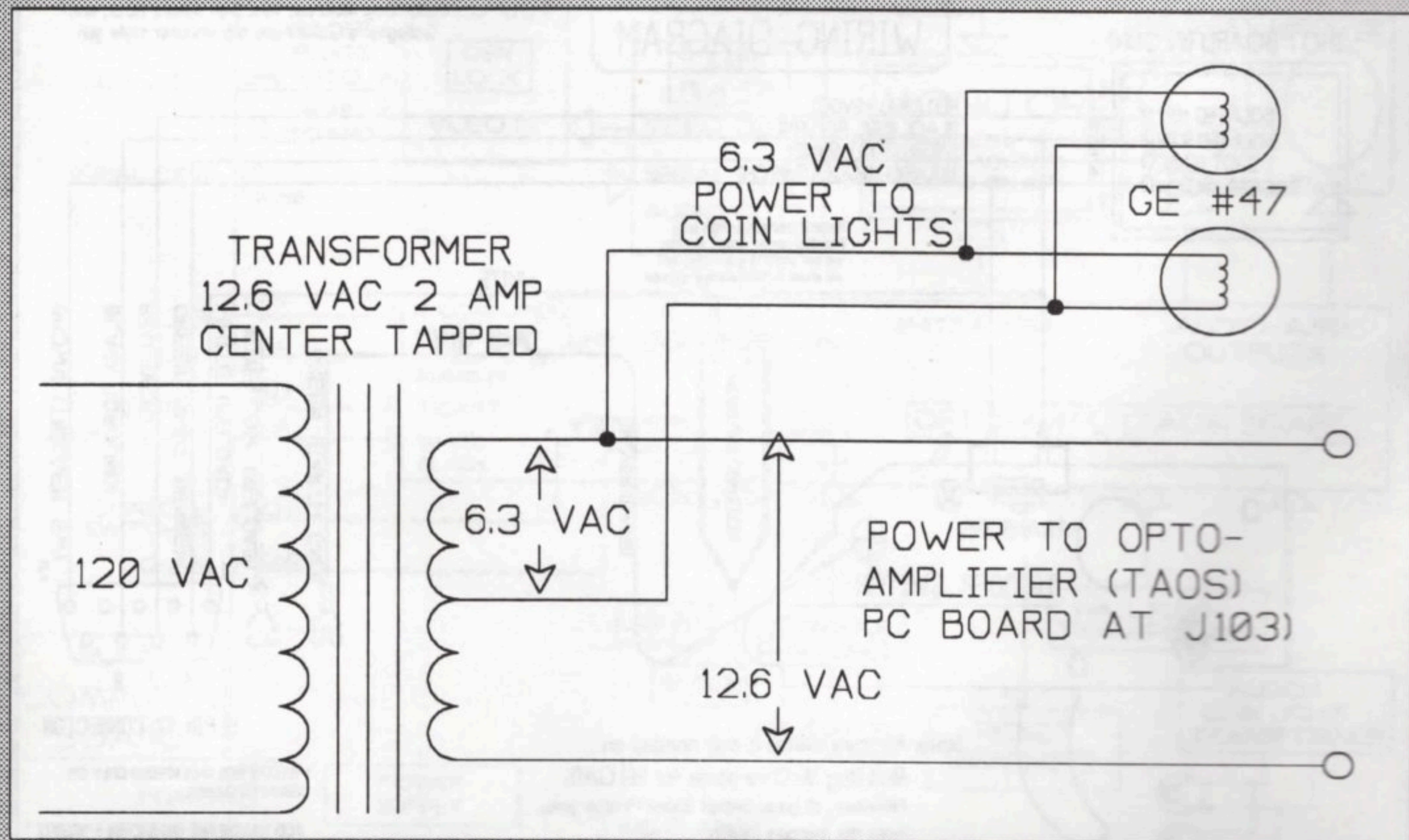
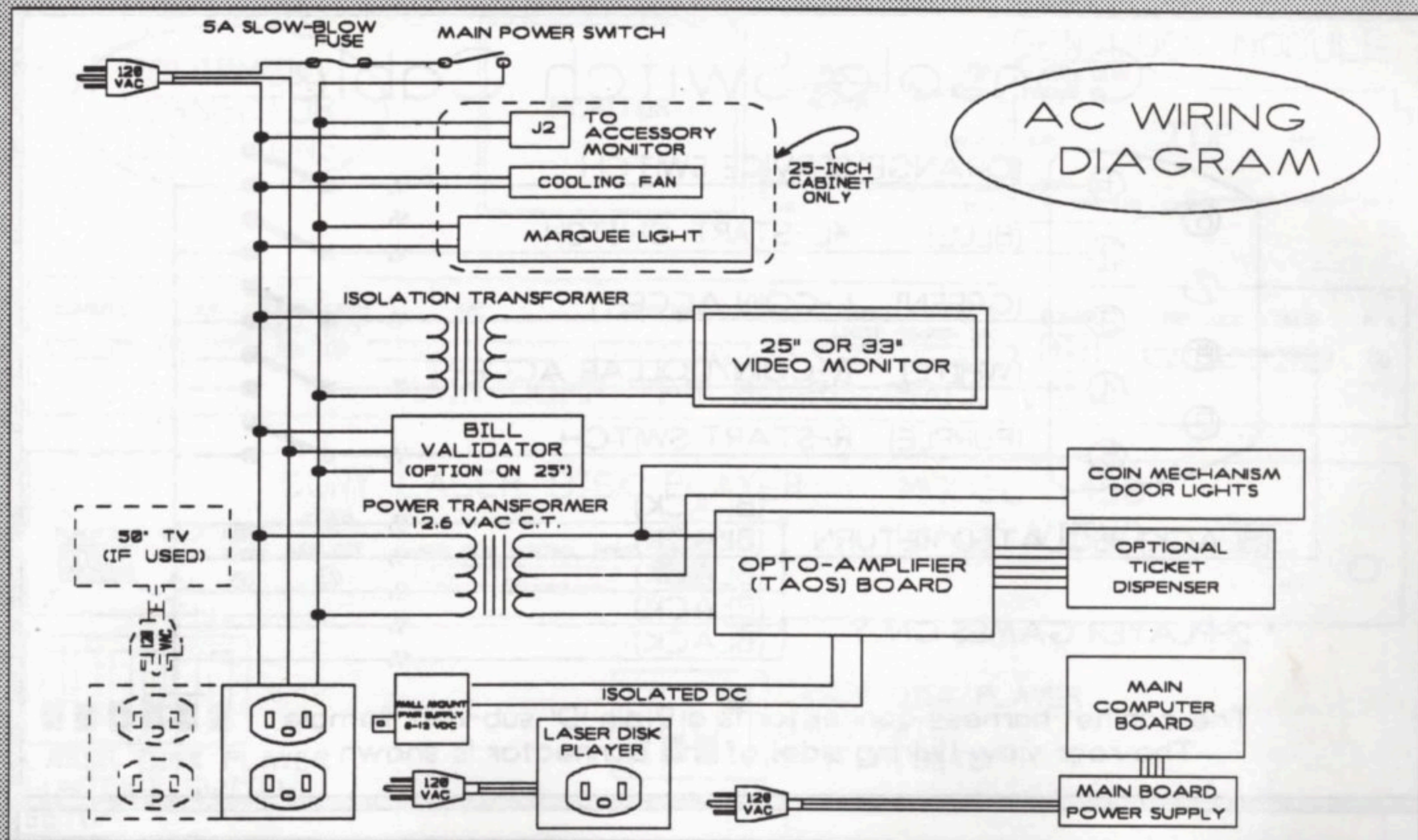
GUN WIRING DIAGRAM

Space Pirates uses a Mauser-style gun.
Mad Dog McCree, Who Shot Johnny Rock, and
Gallagher's Gallery use a 6-shooter style gun

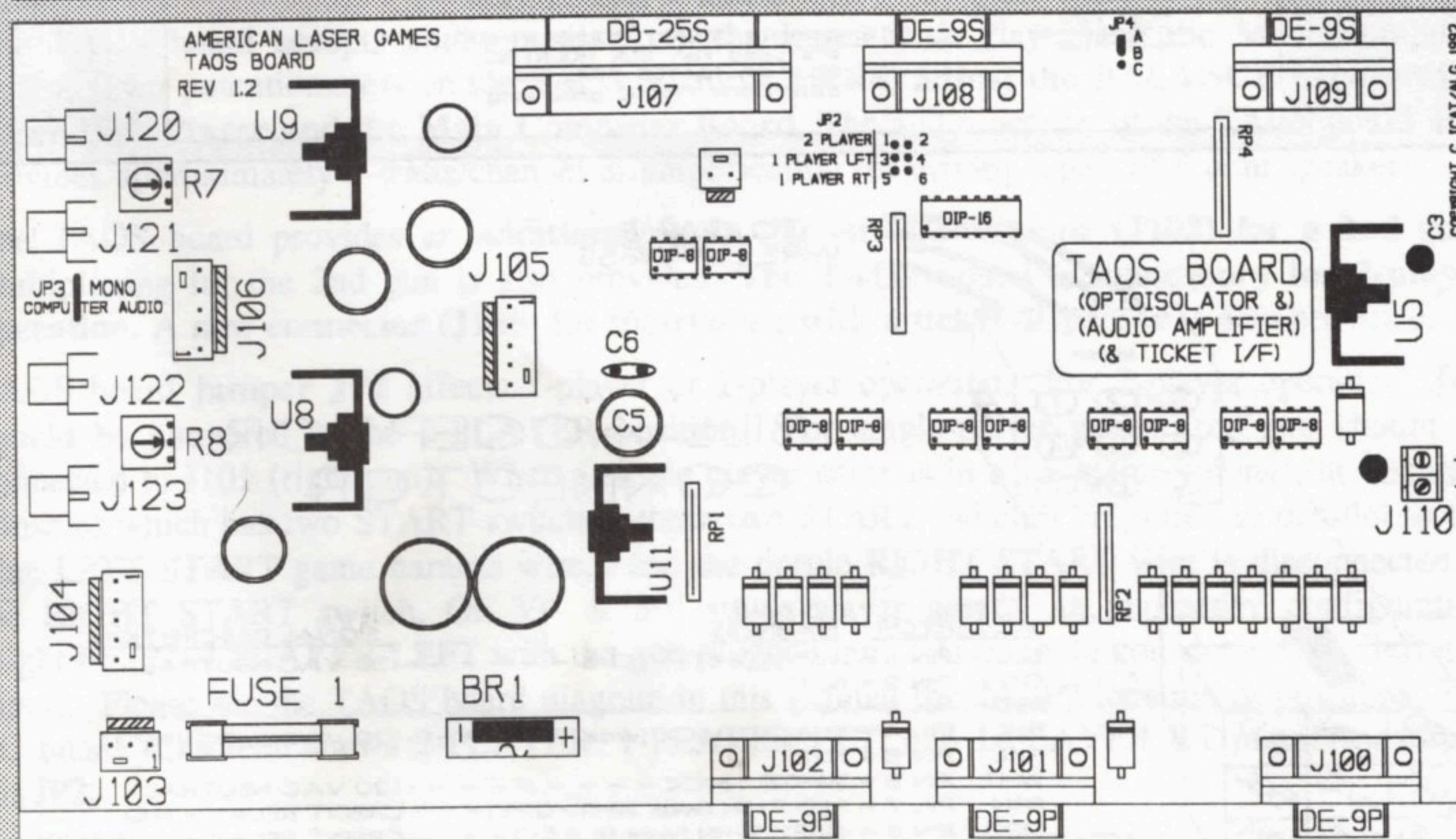
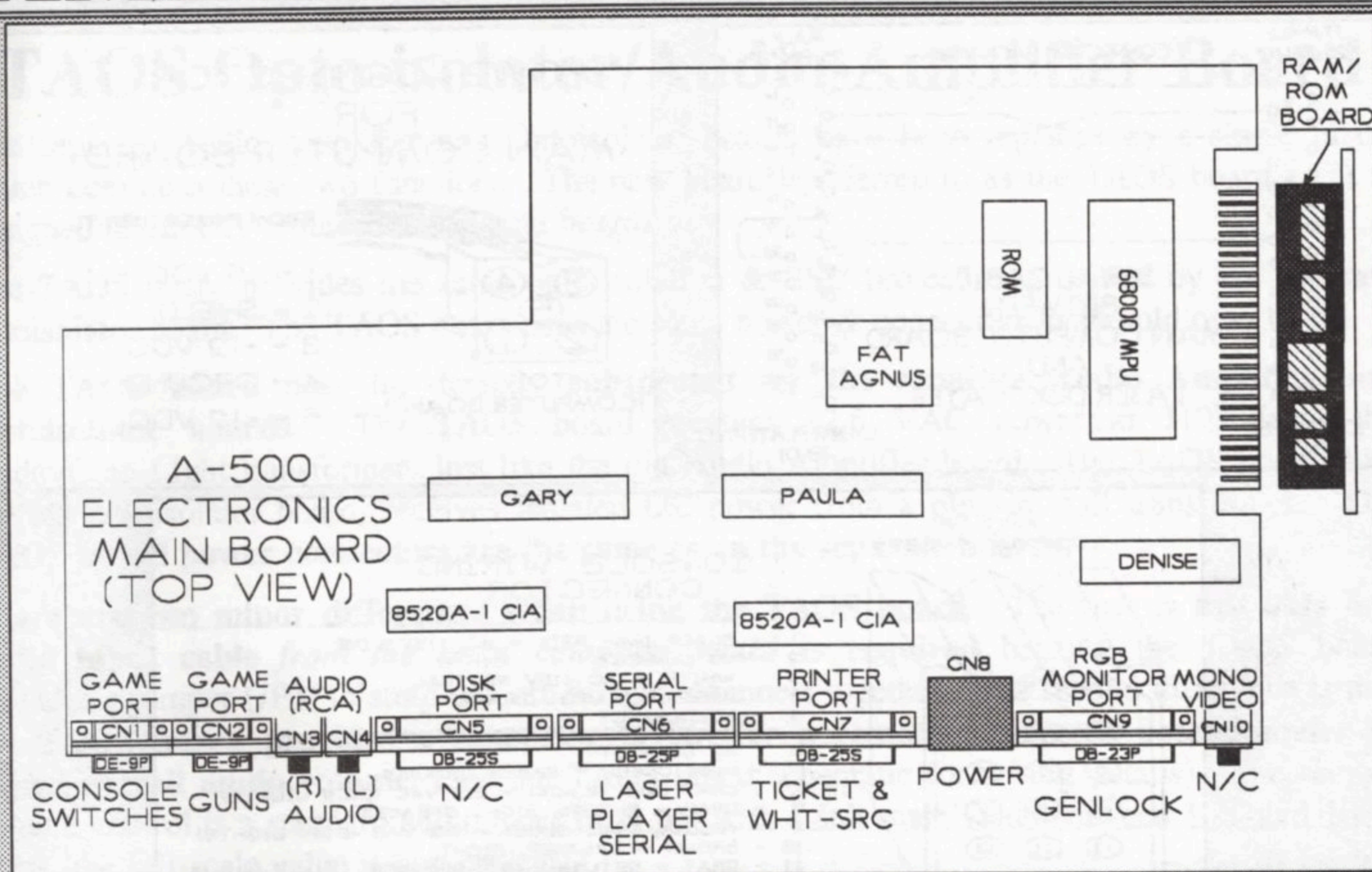
SHOT BOARD (IN GUN)



AMERICAN LASER GAMES

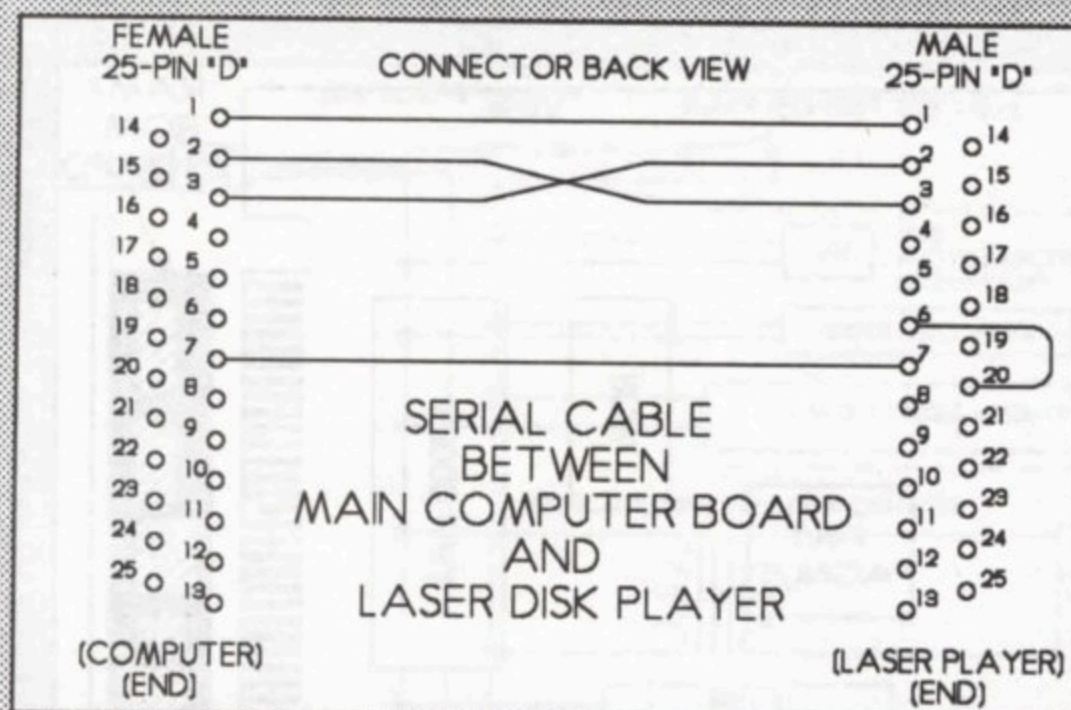


AMERICAN LASER GAMES

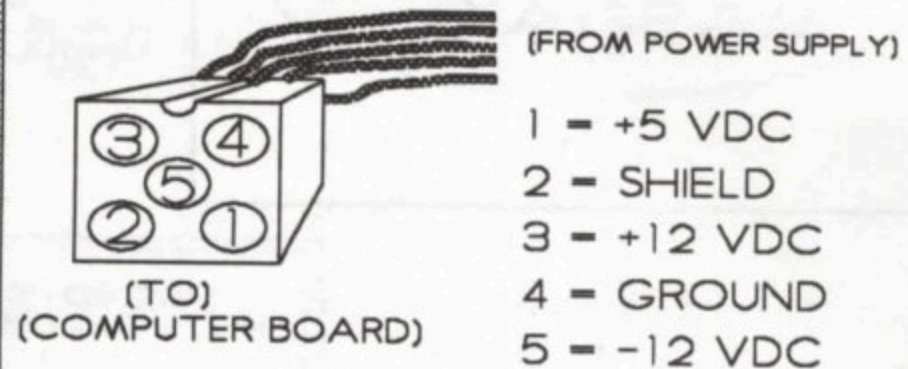


TAOS BOARD (Rev. 1.2) Optoisol/Audio Amplification

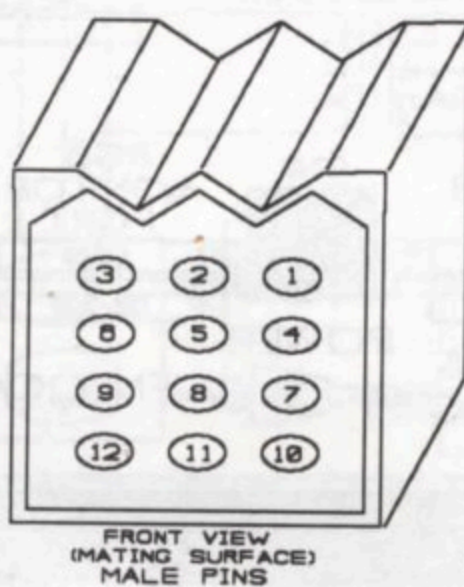
AMERICAN LASER GAMES



POWER CONNECTOR FOR MAIN COMPUTER BOARD



PART OF CONSOLE HARNESS



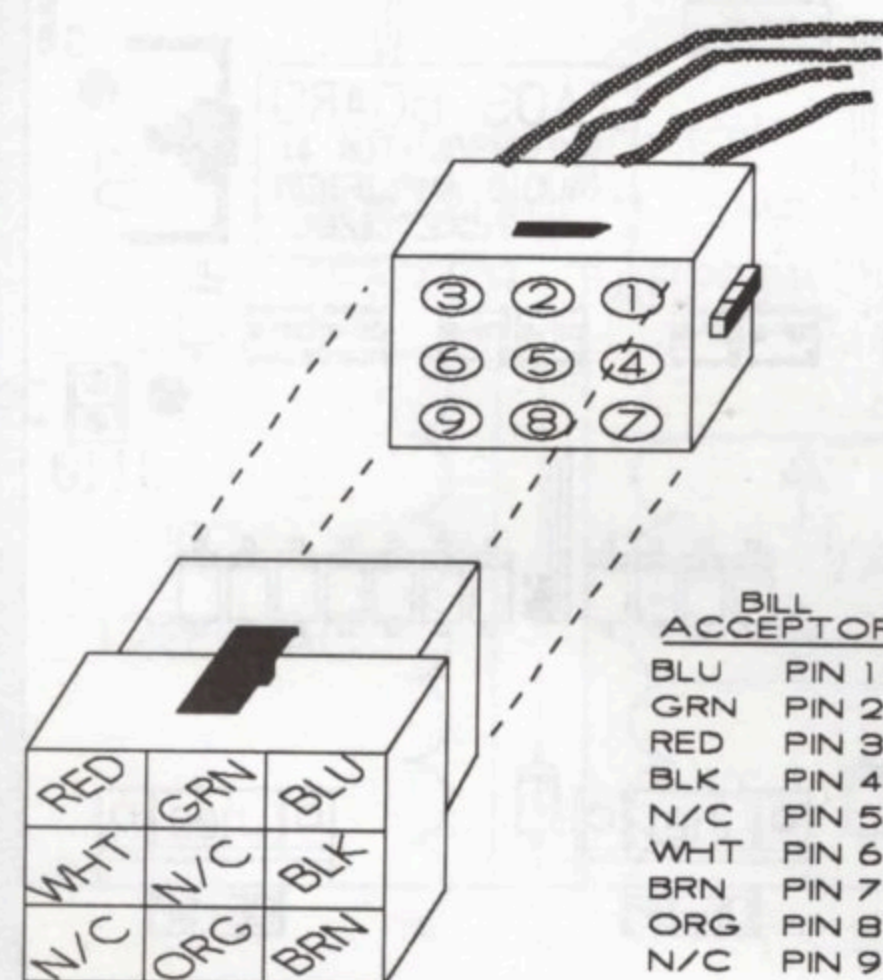
CONSOLE WIRING CONNECTOR

- 1 - BLACK (was RED) - AC 115V LINE
- 2 - NO CONNECT
- 3 - WHITE - AC 115V NEUTRAL
- 4 - GREEN - EARTH GROUND
- 5 - NO CONNECT
- 6 - (was GREEN - EARTH GROUND)
- 7 - GRAY (was BLACK) - 8.3 VAC (COIN LIGHTS)
- 8 - GRAY - RETURN, RIGHT SPEAKER
- 9 - YELLOW (was BLUE) - 8.3 VAC (COIN LIGHTS)
- 10 - BROWN - SPEAKER, RIGHT
- 11 - GRAY - RETURN, LEFT SPEAKER
- 12 - ORANGE - SPEAKER, LEFT

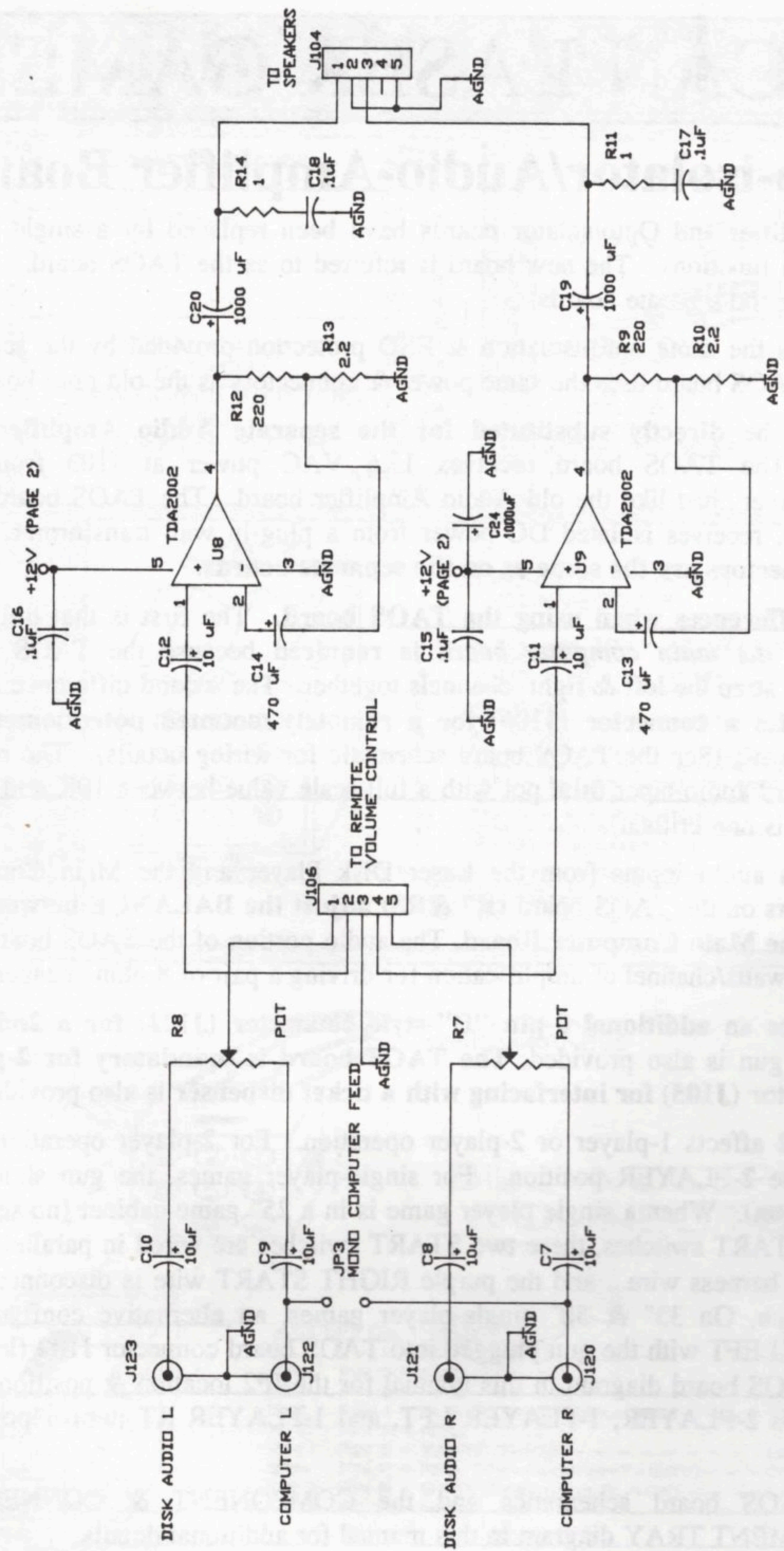
NOTE: Console GREEN GROUND wire(s) are used for: GUN MOUNTING, BILL VALIDATOR BOX, and COIN DOOR FRAME grounding

TO GAME WIRE HARNESS

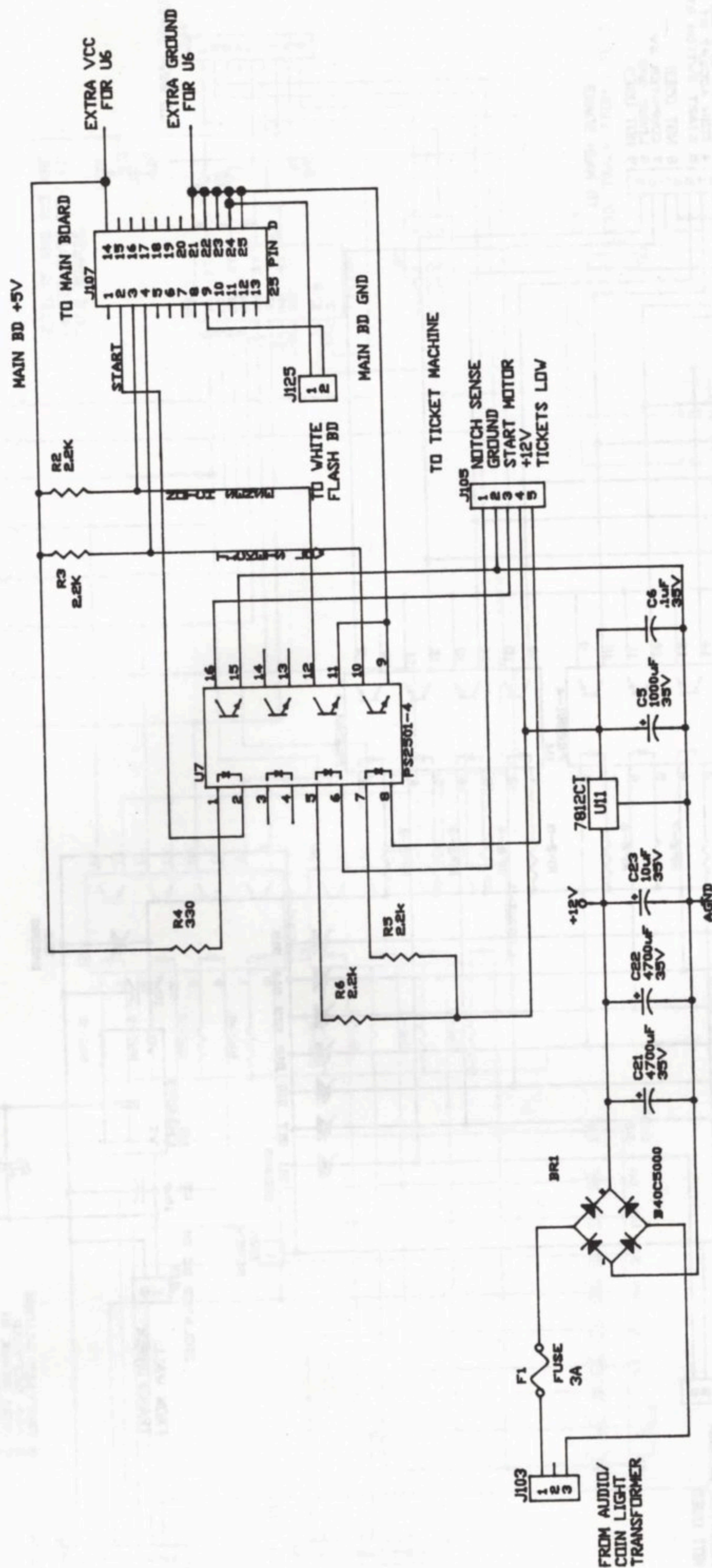
BILL ACCEPTOR CONNECTOR



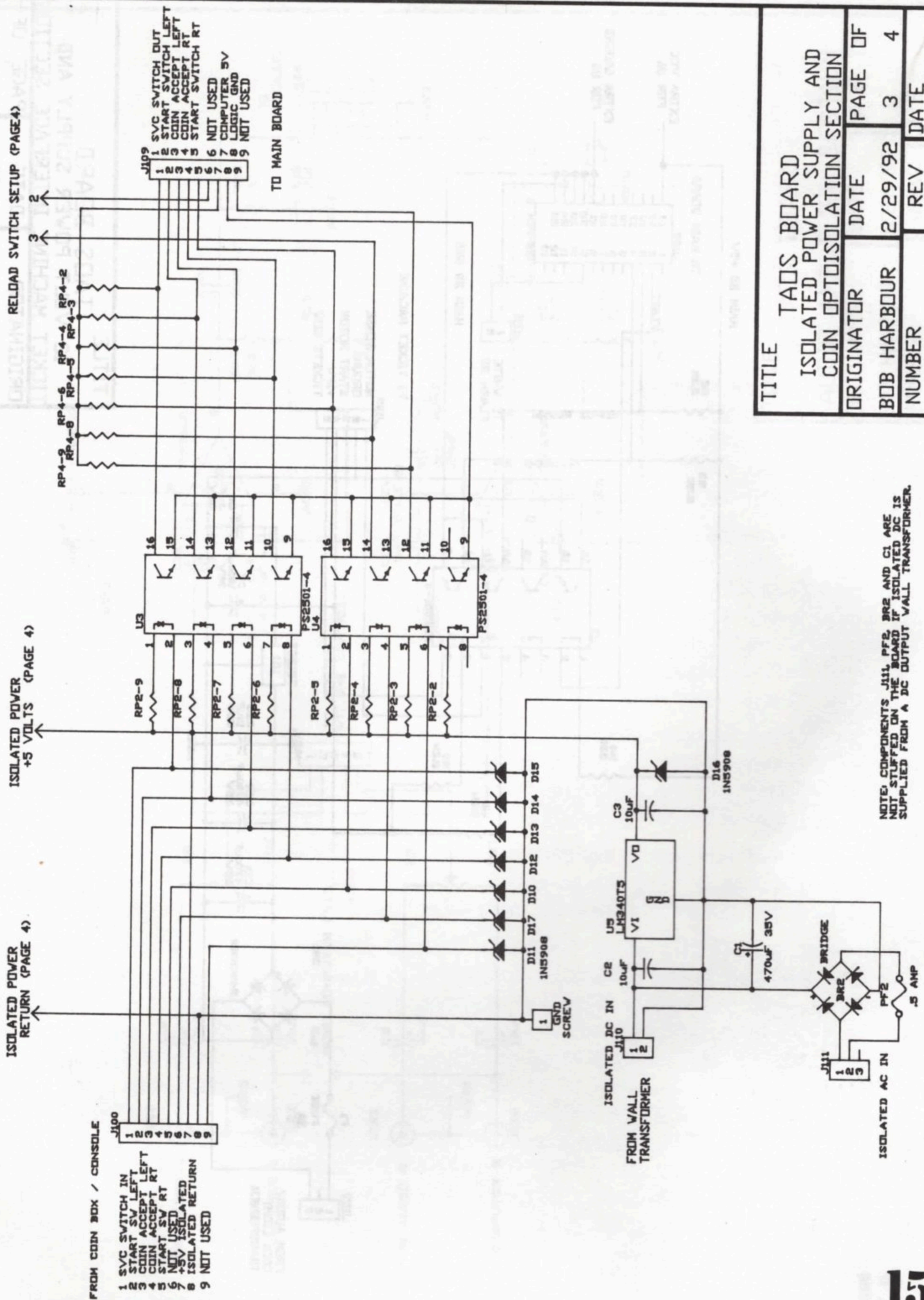
BILL ACCEPTOR	GAME HARNESS	SIGNAL DESCRIPTION
BLU	PIN 1 - JMPR (WHITE) TO 6	- 120 VAC NEUTRAL
GRN	PIN 2 - N/C	
RED	PIN 3 - N/C	
BLK	PIN 4 - BLACK THICK (was RED)	- 120 VAC "HOT" INPUT
N/C	PIN 5 - N/C	
WHT	PIN 6 - WHITE THICK	- 120 VAC NEUTRAL
BRN	PIN 7 - RED THIN (was WHT/GRY)	- CREDIT RELAY - N.O.
ORG	PIN 8 - GREY THIN (was BLACK)	- CREDIT RELAY - COMMON
N/C	PIN 9 - N/C	



TITLE TADS BOARD			
AUDIO AMPLIFIER SECTION			
ORIGINATOR	DATE	PAGE	OF
BOB HARBOUR	2/29/92	1	4
NUMBER	REV	DATE	
	1.2	9/8/92	



TITLE TADS BOARD			
12 VOLT POWER SUPPLY AND			
TICKET MACHINE INTERFACE SECTION			
ORIGINATOR	DATE	PAGE	OF
BOB HARBOUR	2/29/92	2	4
NUMBER	REV	DATE	
	1.2	9/8/92	



TITLE TADS BOARD			
ISOLATED POWER SUPPLY AND			
COIN OPTOISOLATION SECTION			
ORIGINATOR	DATE	PAGE	OF
BOB HARBOUR	2/29/92	3	4
NUMBER	REV	DATE	
	1.2	9/8/92	

J108 LEGEND

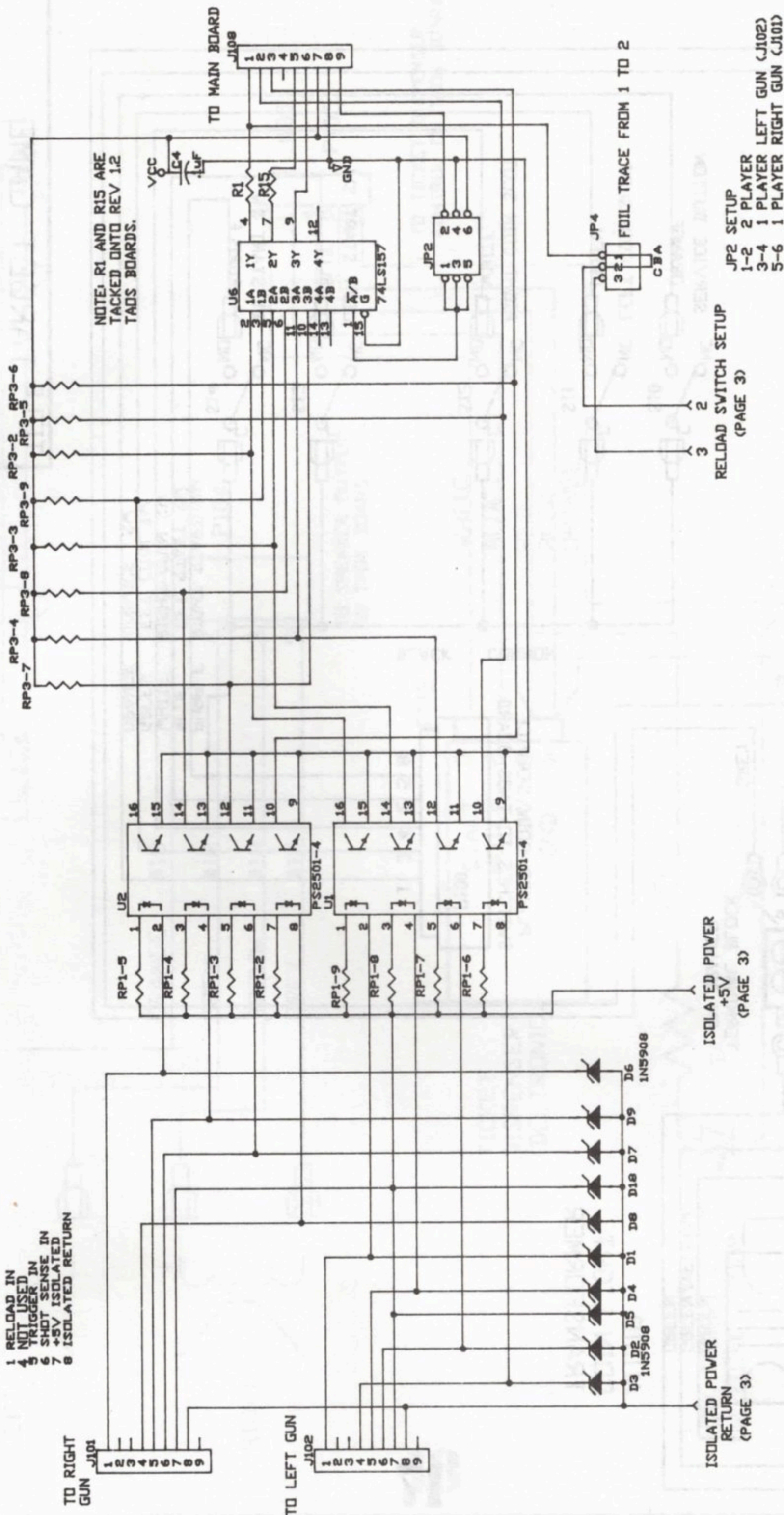
TO GAME BD INPUT
 1 RELOAD OUT
 5 TRIGGER OUT
 6 SHOT SENSE OUT
 8 LOGIC GND
 7 COMPUTER 5V
 9 SELECT A/B

J101, J102 LEGEND TO GUNS

1 RELOAD IN
 4 NOT USED
 5 TRIGGER IN
 6 SHOT SENSE IN
 7 +5V ISOLATED
 8 ISOLATED RETURN

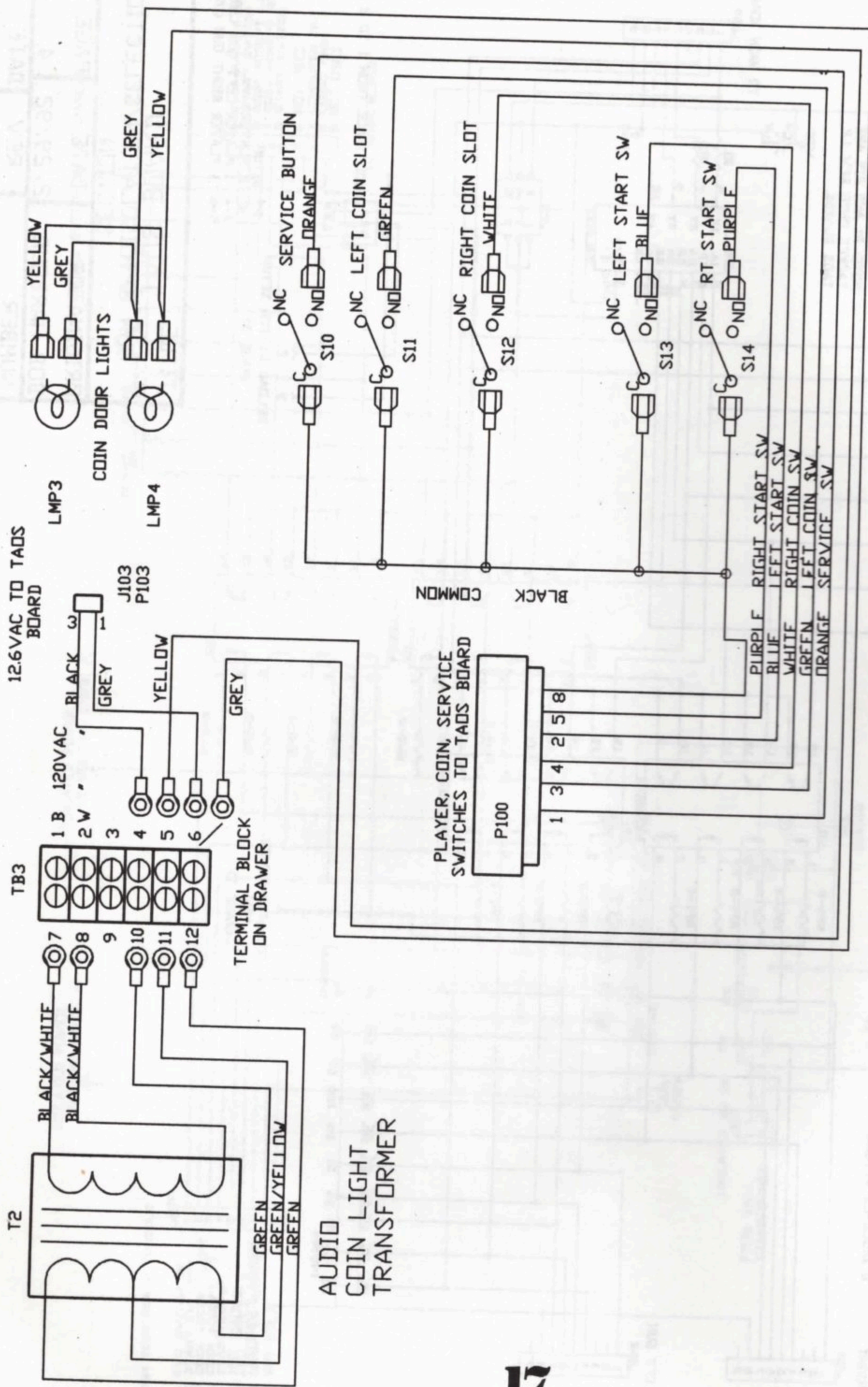
TO RIGHT
 GUN J101

TO LEFT GUN
 J102

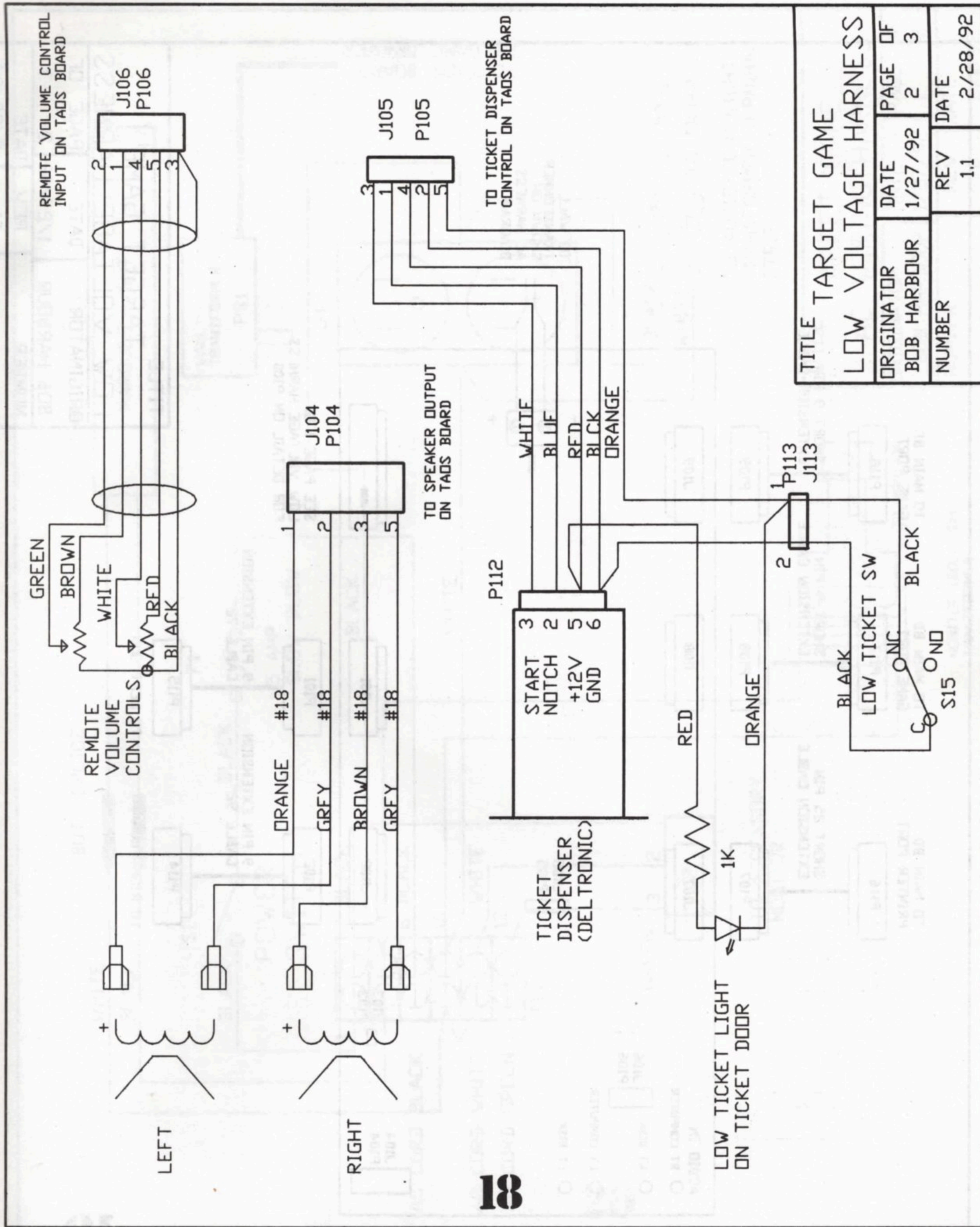


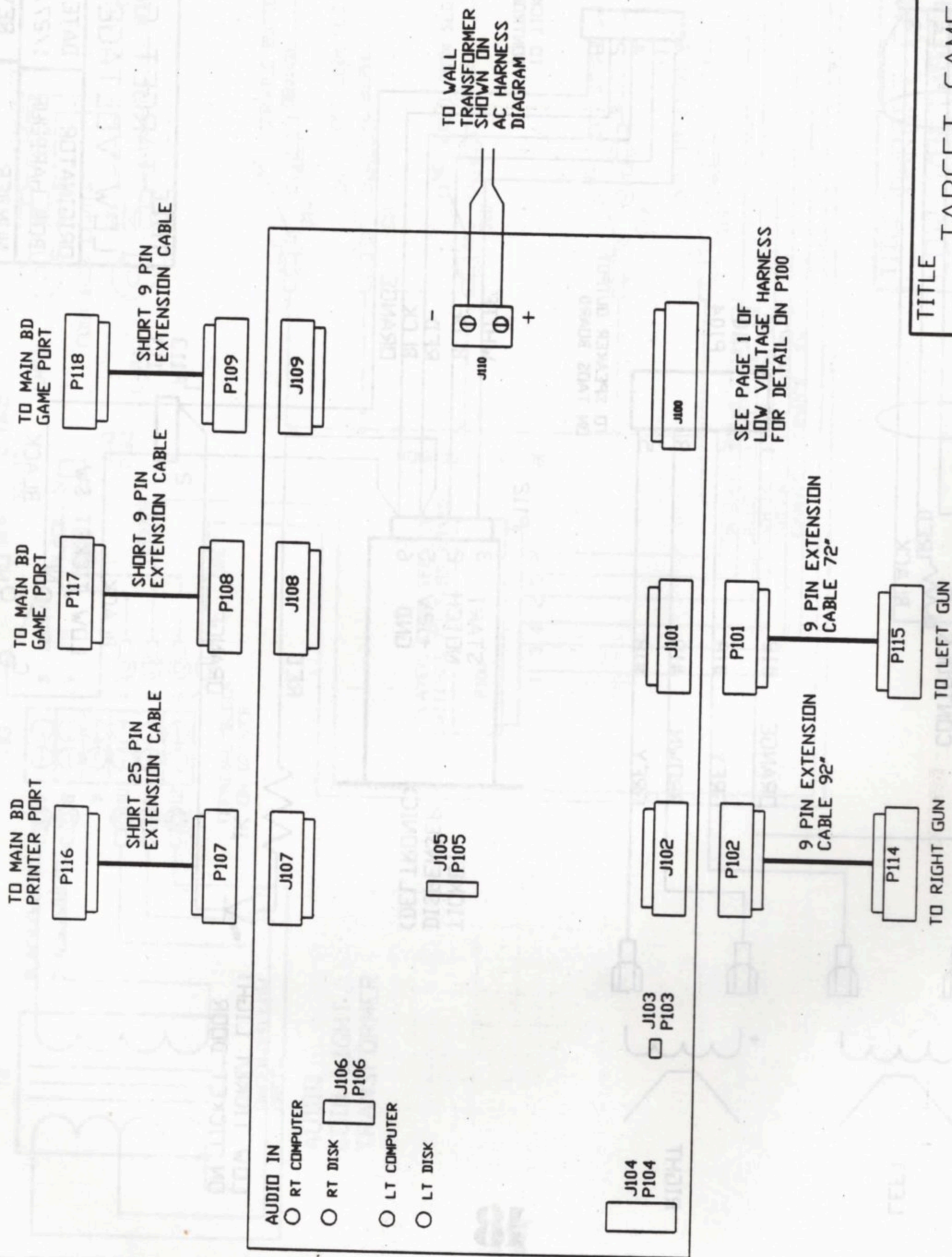
TITLE TADS BOARD
 GUN OPTOISOLATION/SELECTION
 SECTION

ORIGINATOR	DATE	PAGE	OF
BOB HARBOUR	2/29/92	4	4
NUMBER	REV	DATE	
	1.2	9/8/92	

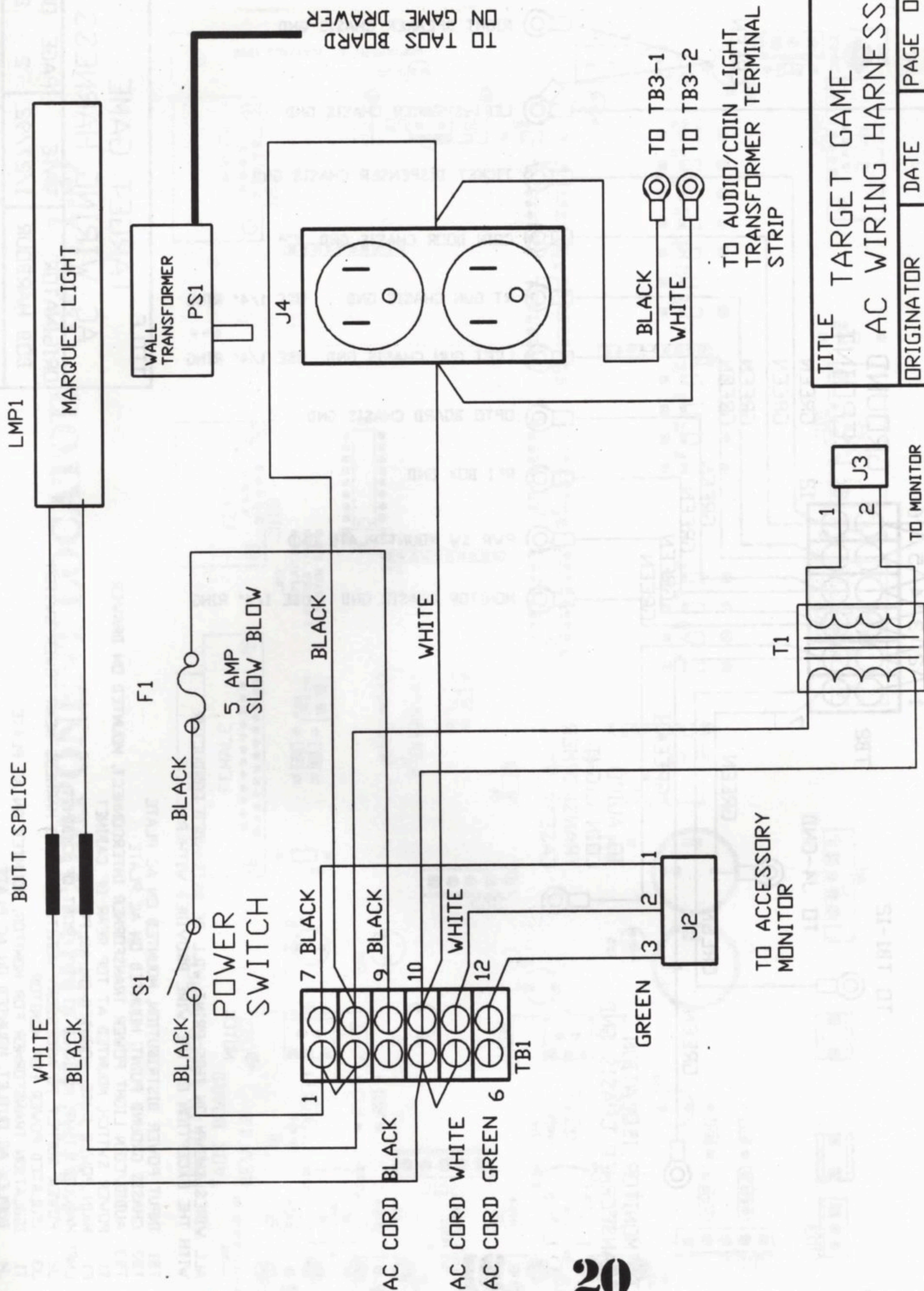


TITLE TARGET GAME			
LOW VOLTAGE HARNESS			
ORIGINATOR	DATE	PAGE	OF
BOB HARBOUR	1/27/92	1	3
NUMBER	REV	DATE	
	1.1	2/28/92	

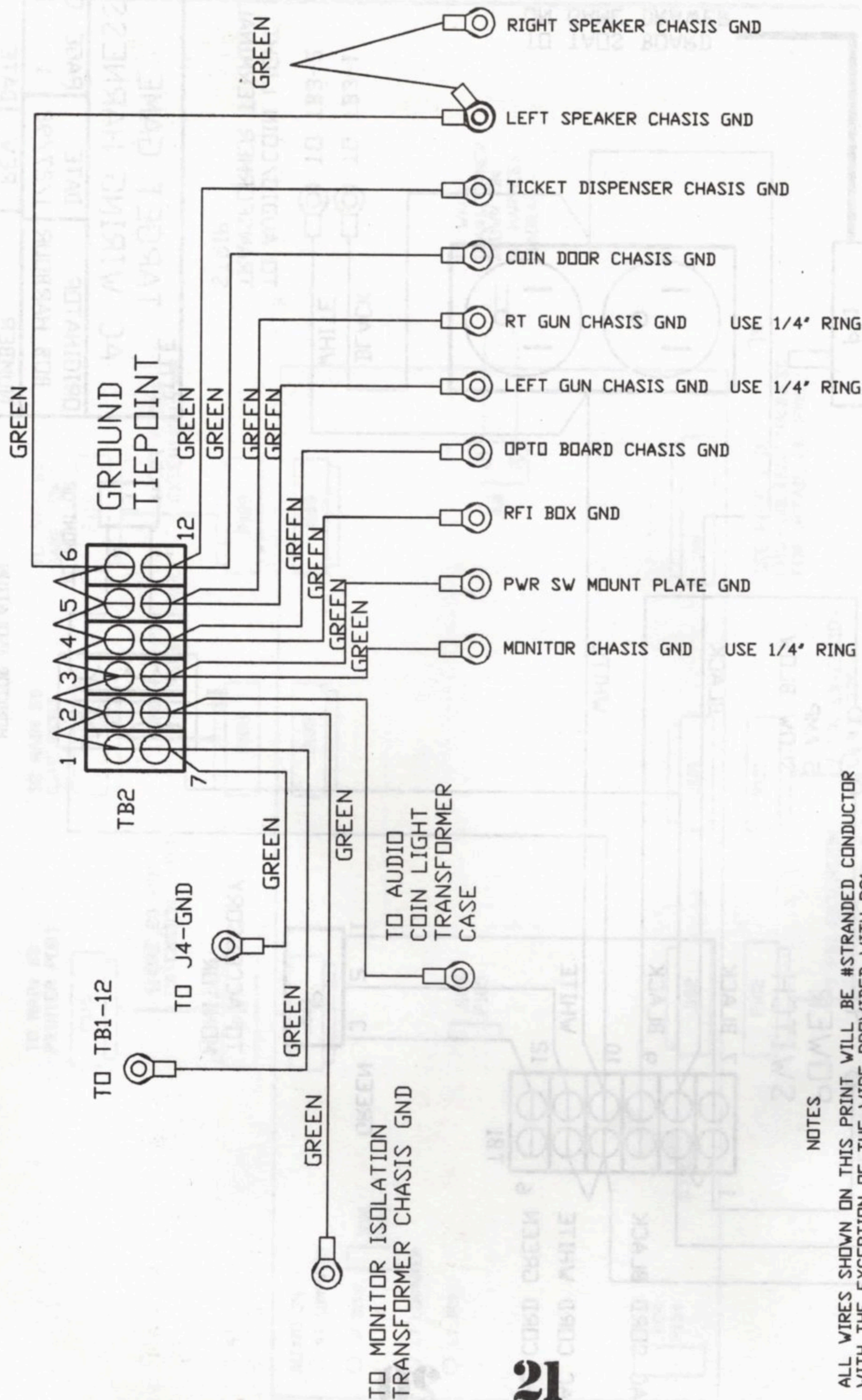




TITLE TARGET GAME			
LOW VOLTAGE HARNESS			
ORIGINATOR	DATE	PAGE	OF
BOB HARBOUR	1/27/92	3	3
NUMBER	REV	DATE	
	1.1	2/28/92	



TITLE TARGET GAME			
AC WIRING HARNESS			
ORIGINATOR	DATE	PAGE	OF
BOB HARBOUR	1/27/92	1	2
NUMBER	REV	DATE	
	1.0	2/1/92	



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NOTES

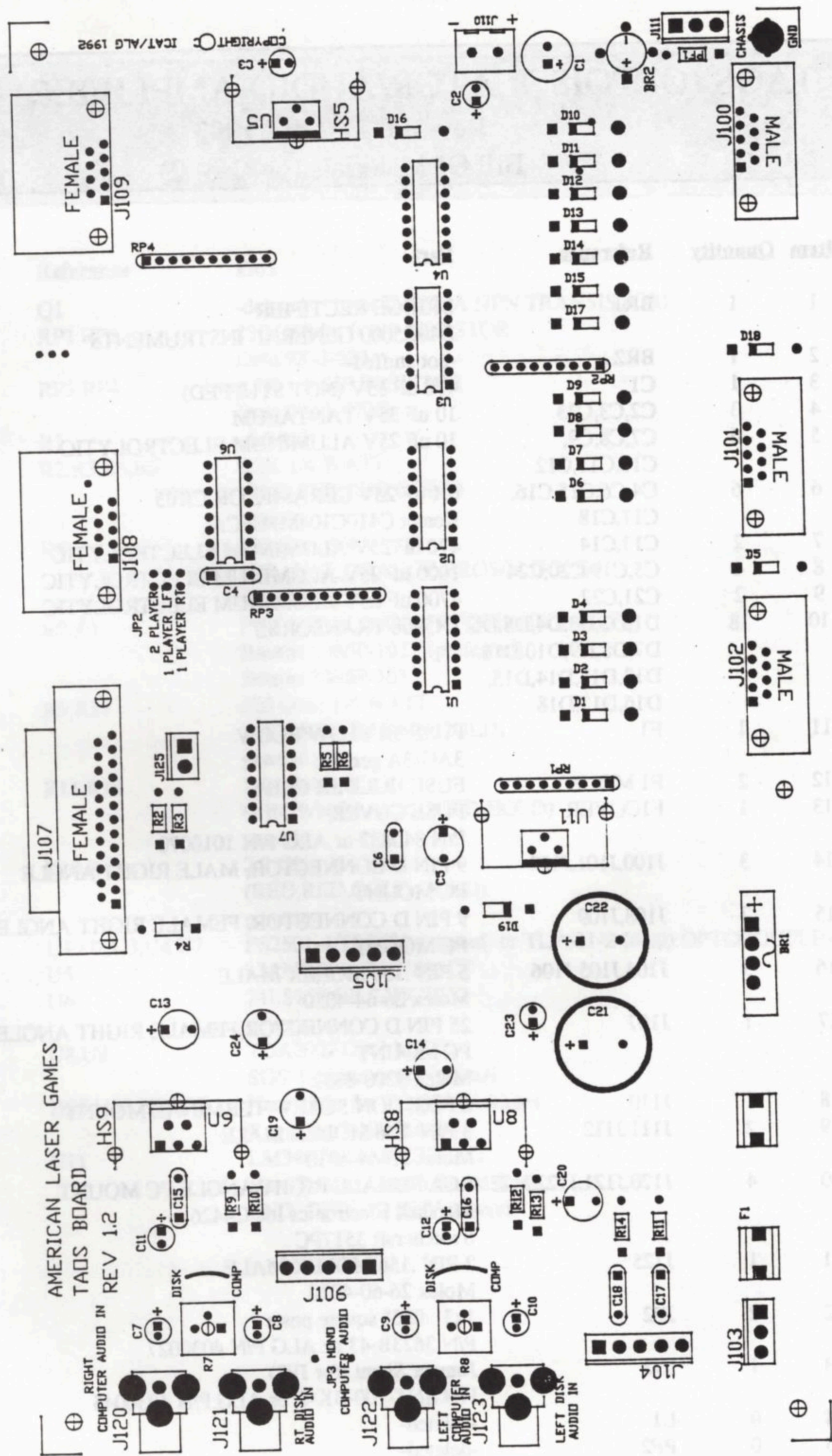
ALL WIRES SHOWN ON THIS PRINT WILL BE #18 STRANDED CONDUCTOR WITH THE EXCEPTION OF THE WIRE PROVIDED WITH PS1

- TB1 INPUT POWER DISTRIBUTION, MOUNTED ON AC PLATE
- TB2 CHASIS GROUND POINT, MOUNTED ON AC PLATE
- TB3 AUDIO/COIN LIGHT POWER TRANSFORMER INTERCONNECT, MOUNTED ON DRAWER
- S1 POWER SWITCH, MOUNTED AT TOP REAR OF CABINET
- F1 MAIN POWER FUSE, MOUNTED ON AC PLATE
- LMP1 MARQUIS LIGHT, MOUNTED AT TOP FRONT OF CABINET
- J2 POWER SUPPLY TO ACCESSORY MONITOR, TIE TO HARNESS NEAR POWER SWITCH
- J3 ISOLATED POWER FOR MONITOR
- T1 ISOLATION TRANSFORMER FOR MONITOR, MOUNTED ON AC PLATE
- J4 DUPLEX AC OUTLET, MOUNTED ON AC PLATE
- PS1 WALL PLUG TRANSFORMER STANCOR W120DUJ50-1 ,PLUGS INTO J4

TITLE TARGET GAME AC WIRING HARNESS

ORIGINATOR	DATE	PAGE	OF
BOB HARBOUR	1/27/92	2	2
NUMBER	REV	DATE	
	1.0	2/1/92	

TAOS BOARD COMPONENT LOCATOR (Rev. 1.2)



TAOS (OPTOISOLATOR/AUDIO AMPLIFIER) BOARD

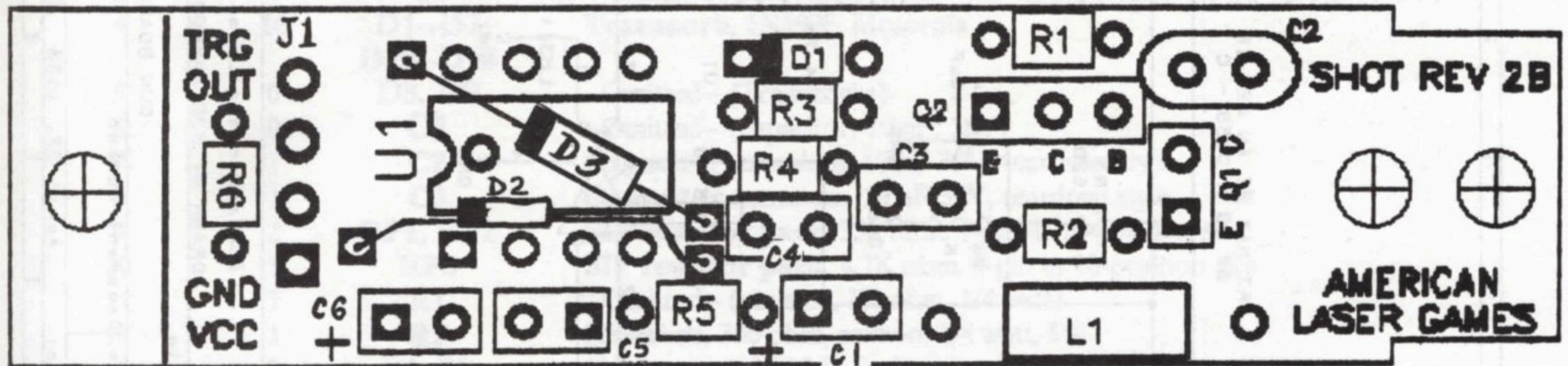
Revised: 27 April 1993

Bill Of Materials (pg. 1 of 2)

TAOS Rev. 1.2

Item	Quantity	Reference	Part
1	1	BR1	BRIDGE RECTIFIER B40C5000 GENERAL INSTRUMENTS
2	1	BR2	-not stuffed-
3	1	C1	470 uF 35V (NOT STUFFED)
4	3	C2,C3,C23	10 uF 35V TANTALUM
5	6	C7,C8,C9, C10,C11,C12	10 uF 25V ALUMINUM ELECTROLYTIC
6	6	C4,C6,C15,C16, C17,C18	0.1 uF 25V CERAMIC OR CK05 Kemet C410C104M5U5CA
7	2	C13,C14	470 uF 25V ALUMINUM ELECTROLYTIC
8	4	C5,C19,C20,C24	1000 uF 25V ALUMINUM ELECTROLYTIC
9	2	C21,C22	4700 uF 25V ALUMINUM ELECTROLYTIC
10	18	D1,D2,D3,D4,D5,D6, D7,D8,D9,D10,D11, D12,D13,D14,D15, D16,D17,D18	1N5908 TRANSORBS
11	1	F1	FUSE 3A SLOW BLOW 3AG-3A generic
12	2	F1 MOUNT	FUSE HOLDER CLIPS
13	1	F1COVER	FUSE COVER P/N 840832 or ALG P/N 1010070
14	3	J100,J101,J102	9 PIN D CONNECTOR, MALE RIGHT ANGLE PC MOUNT
15	2	J108,J109	9 PIN D CONNECTOR, FEMALE RIGHT ANGLE PC MOUNT
16	3	J104,J105,J106	5 PIN .156 MOLEX MALE Molex 26-64-4050
17	1	J107	25 PIN D CONNECTOR,FEMALE RIGHT ANGLE PC MOUNT Molex 8200-8352
18	1	J110	2 POSITION SCREW TERMINAL MOUNT
19	2	J111,J112	3 PIN .156 MOLEX MALE Molex 26-64-4030
20	4	J120,J121,J122,J123	RCA FEMALE RIGHT ANGLE PC MOUNT Marshall Electronics Inc C4426 Switchcraft 3517PC
21	1	J125	2 PIN .156 MOLEX MALE Molex 26-60-4020
22	2	JP2	1x3 .025" square posts P/N 3623B-43 or ALG P/N 4030027
23	1	JS2	Jumper Shunt (for JP2) P/N SNT-100-BK-G or ALG P/N 4030045
24	0	L1	-deleted-
25	0	PF2	-deleted-

GUN SHOT BOARD COMPONENT LOCATOR



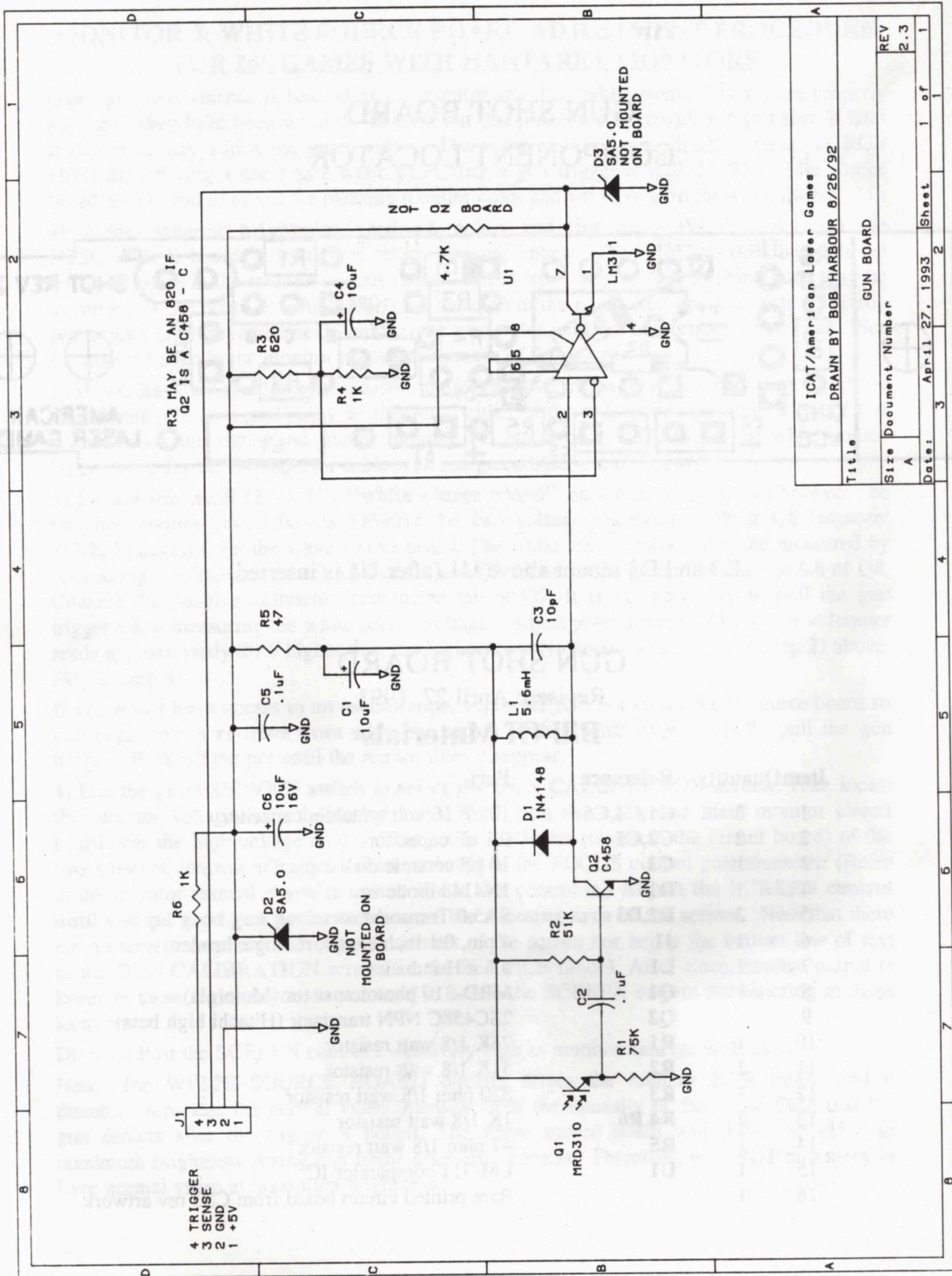
D2 and D3 mount above U1 (after U1 is inserted)

GUN SHOT BOARD

Revised: April 27, 1993

Bill Of Materials

Item	Quantity	Reference	Part
1	3	C1,C4,C6	10 uF 16 volt tantalum capacitor
2	2	C2,C5	0.1 uF capacitor
3	1	C3	10 pF ceramic disk capacitor
4	1	D1	1N4148 diode
5	2	D2,D3	SA5.0 Transorbs
6	1	J1	4 pin, 0.1 inch center, rt. angle header
7	1	L1	5.6 mH inductor
8	1	Q1	MRD-310 phototransistor (Motorola)
9	1	Q2	2SC458C NPN transistor (Hitachi high beta)
10	1	R1	75K 1/8 watt resistor
11	1	R2	51K 1/8 watt resistor
12	1	R3	820 ohm 1/8 watt resistor
13	2	R4,R6	1K 1/8 watt resistor
14	1	R5	47 ohm 1/8 watt resistor
15	1	U1	LM-311 comparator IC
16	1		Bare printed circuit board from CS3 rev artwork

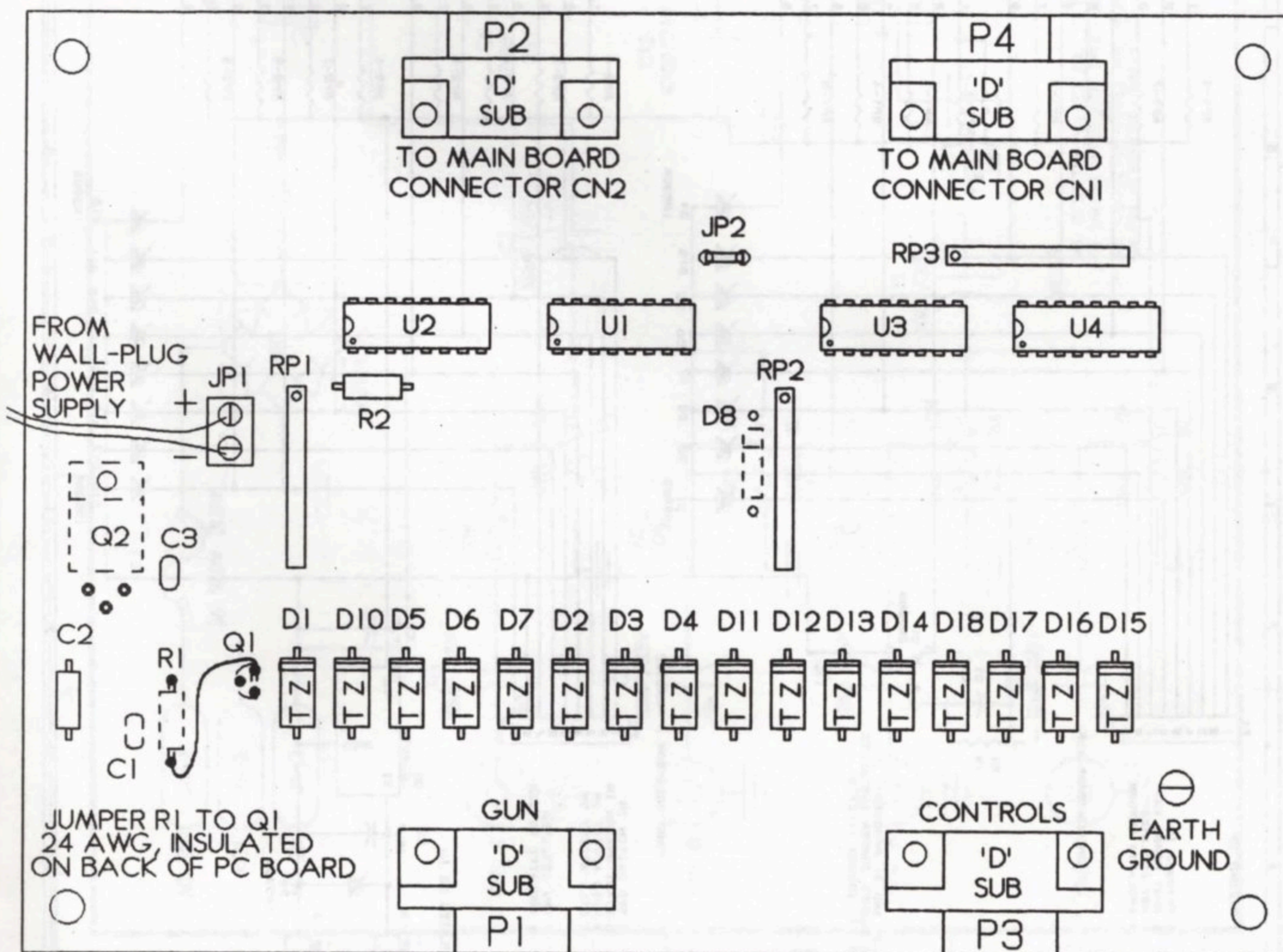


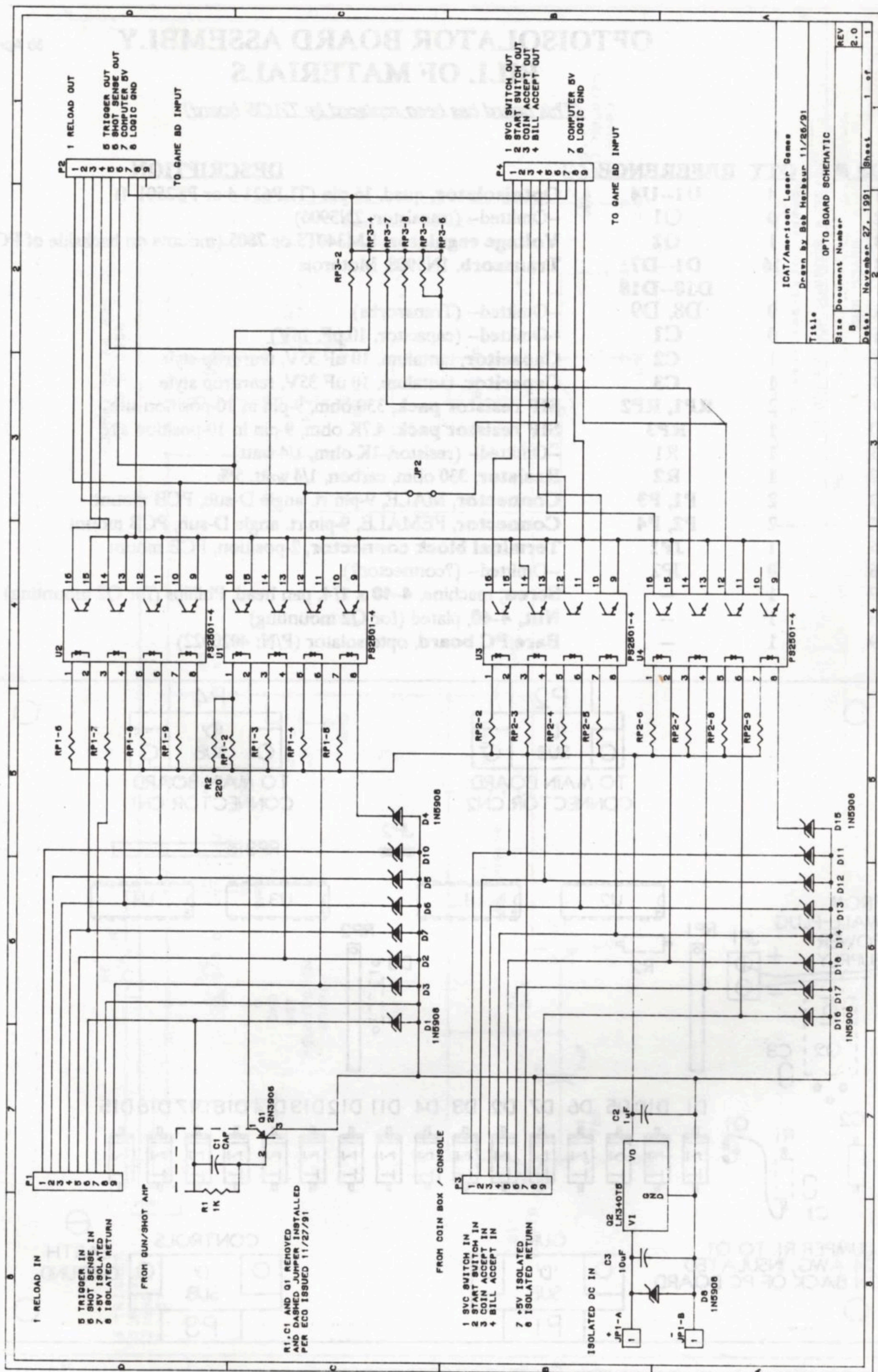
OPTOISOLATOR BOARD ASSEMBLY BILL OF MATERIALS

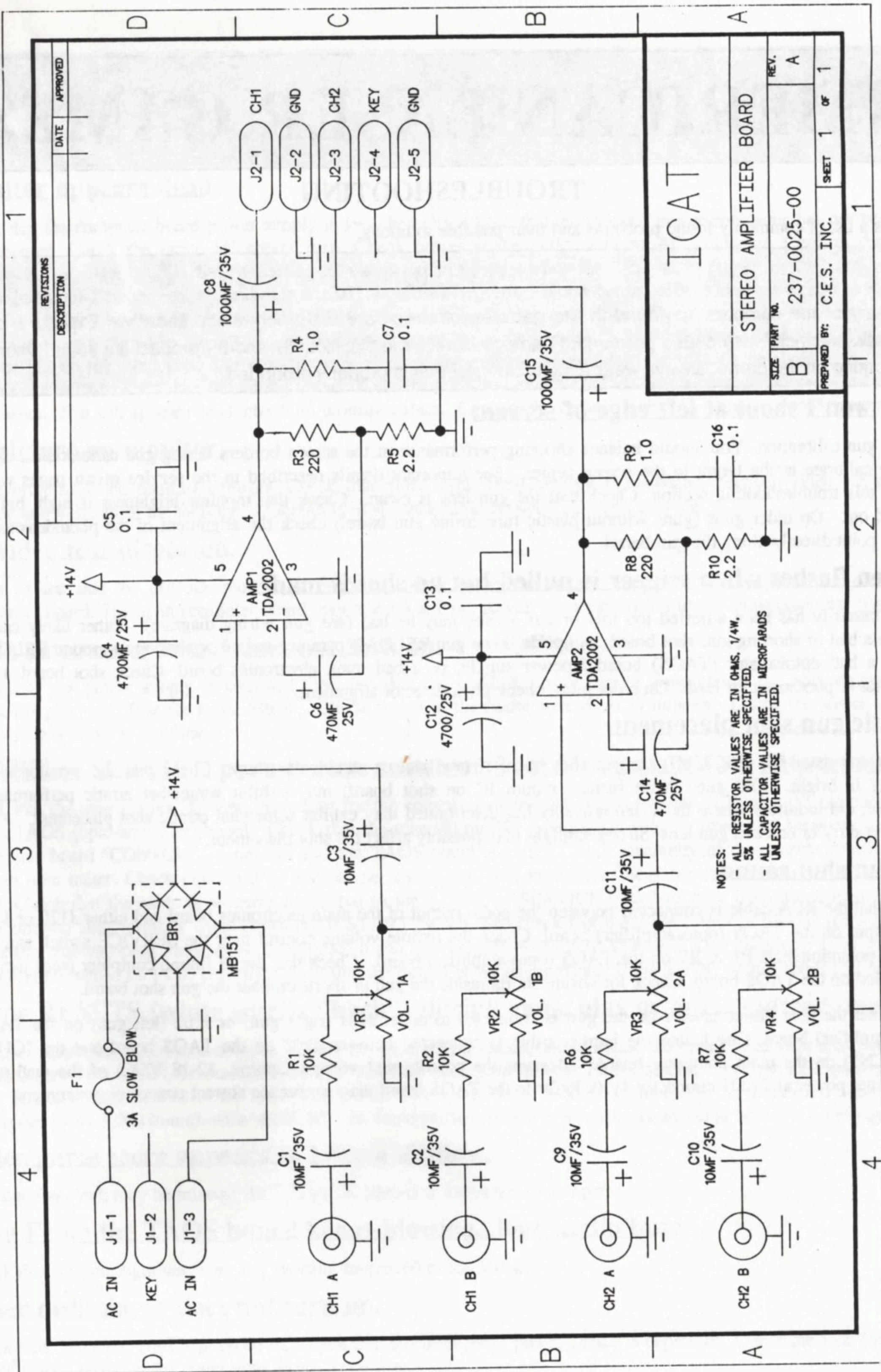
30 April 1993

(This board has been replaced by TAOS board)

ITEM #	QTY	REFERENCE	DESCRIPTION
1	4	U1--U4	Optoisolator, quad, 16-pin (TLP621-4 or PS2501-4)
2	0	Q1	--Omitted-- (transistor, 2N3906)
3	1	Q2	Voltage regulator, LM340T5 or 7805 (mounts on backside of PCB)
4	16	D1--D7 D10--D18	Transzorb, 1N5908, Motorola
5	0	D8, D9	--Omitted-- (Transzorbs)
6	0	C1	--Omitted-- (capacitor, 10 pF, 16V)
7	1	C2	Capacitor, tantalum, 10 uF 35V, teardrop style
8	1	C3	Capacitor, tantalum, 10 uF 35V, teardrop style
9	2	RP1, RP2	SIP resistor pack, 330 ohm, 9-pin in 10-position site
10	1	RP3	SIP resistor pack, 4.7K ohm, 9-pin in 10-position site
11	1	R1	--Omitted-- (resistor, 1K ohm, 1/4 watt)
12	1	R2	Resistor, 330 ohm, carbon, 1/4 watt, 5%
13	2	P1, P3	Connector, MALE, 9-pin rt. angle D-sub, PCB mount
14	2	P2, P4	Connector, FEMALE, 9-pin rt. angle D-sub, PCB mount
15	1	JP1	Terminal block connector, 2-position, PCB mount
16	0	JP2	--Omitted-- (?connector?)
17	1	--	Screw, machine, 4-40 x 1/4, pan head, Phillips (for Q2 mounting)
18	1	--	Nut, 4-40, plated (for Q2 mounting)
19	1	--	Bare PC board, optoisolator (P/N: 4020022)







ICAT

STEREO AMPLIFIER BOARD

SIZE	PART NO.	REV.
B	237-0025-00	A

PREPARED BY: C.E.S. INC. SHEET 1 OF 1